



Ayahuasca, DMT, and Mental Health: A Current Review of Scientific Studies

Dráulio B. Araújo^{1,2} · Lucas O. Maia^{1,2} · Tiago Arruda-Sanchez^{1,3} · Nicole Galvão-Coelho^{1,4} ·
Fernanda Palhano-Fontes^{1,2}

Received: 12 May 2025 / Accepted: 27 January 2026
© The Author(s) 2026

Abstract

Purpose of Review Summarizes preclinical and clinical evidence on ayahuasca—a traditional Amazonian brew combining *N, N*-Dimethyltryptamine (DMT) and β -carbolines—in treating depression, anxiety, post-traumatic stress disorder, substance use, eating, and personality disorders.

Recent Findings Preclinical work shows enhanced neuroplasticity, reduced inflammation, and oxidative stress. Human neuroimaging reveals decreased default mode network activity, increased functional connectivity and brain entropy, indicating a flexible neural state and modulation of pathways related to neuroplasticity, inflammation, and stress response.

Summary While promising, the evidence is mainly observational. Users report emotional breakthroughs, increased self-awareness, and mystical experiences tied to therapeutic outcomes. Ayahuasca appears to target core psychological and neurobiological processes across disorders but requires caution in psychotic or bipolar individuals and should be administered with support. Randomized trials are needed to confirm efficacy and safety. In structured, culturally sensitive settings, ayahuasca/DMT may offer a novel treatment and, by integrating indigenous wisdom with modern science, deepen our understanding of healing and consciousness.

Keywords Ayahuasca · *N, N*-DMT · Mental health · Depression · PTSD · Substance use disorder

Introduction

Humans have used psychedelic substances across diverse cultures and historical periods. Archaeological evidence indicates that ritualistic psychedelic use dates back at least 9000 years. These substances are potent serotonin agonists, inducing altered states of consciousness often compared to dreams [1]. Ayahuasca, a traditional psychoactive

brew from the Amazon, contains *N, N*-Dimethyltryptamine (DMT), a naturally occurring indole alkaloid found in many plants and animals.

The term “ayahuasca” comes from Quechua, meaning “vine of the spirits”, and it has long been used in spiritual and healing rituals by Indigenous Amazonian peoples [2, 3]. The brew is made by boiling *Banisteriopsis caapi*, which contains β -carboline monoamine oxidase-A (MAO-A) inhibitors, with *Psychotria viridis*, which contains DMT, to yield a concentrated decoction [4]. This combination allows DMT to bypass enzymatic degradation and exert psychoactive effects when ingested orally [5]. The acute experience typically lasts 4–6 h and includes complex visual phenomena, increased introspection, altered perception, nausea, vomiting, and elevated salivary cortisol [6–8]. Users frequently report vivid mental imagery, self-transcendence, and insights, often described as transformative and healing [7, 9].

The first Western scientific encounter with ayahuasca dates back to 1851, when botanist Richard Spruce led an expedition to the Amazon. He documented the use of the

✉ Dráulio B. Araújo
draulio@neuro.ufrn.br

¹ Center for Advanced Medical Psychedelics (CAMP), Natal, RN, Brazil

² Brain Institute, Federal University of Rio Grande do Norte, Av. Senador Salgado Filho, 3000, Natal, RN 59078-900, Brazil

³ Institute of Psychiatry, Medical School, Federal University of Rio de Janeiro, Rio de Janeiro, RJ, Brazil

⁴ Center for Biosciences, Federal University of Rio Grande do Norte, Natal, RN, Brazil

caapi vine in ceremonial contexts and noted its intense psychoactive effects, which he compared to those of opium and hemp [10].

By the early 20th century, the use of ayahuasca had expanded to urban areas through syncretic religious movements, such as Santo Daime and the União do Vegetal (UDV). In 1987, the Brazilian government officially recognized its religious use, paving the way for regulated ceremonial practices and international dissemination [11]. As its visibility increased globally, interest among scientists

grew, especially concerning its therapeutic potential for psychiatric conditions [3]. In fact, by 1990, the scientific research into ayahuasca began gaining momentum. Early observational studies suggested that ayahuasca could alleviate symptoms of depression, anxiety, post-traumatic stress disorder (PTSD), and substance use disorders (SUD) [12, 13]. Table 1 summarizes preclinical and clinical evidence of ayahuasca and DMT in psychiatric disorders.

Ayahuasca shows a favorable safety profile when used in structured contexts [14]. The Hoasca Project, a long-term

Table 1 Summary of preclinical and clinical evidence of Ayahuasca in psychiatric disorders

Disorder	Evidence Type	Model/Study Design	Sample/Species	Intervention (Dose, Regimen)	Comparator/Control Group	Main Outcomes	Assessment Tools/Scales
Substance Use Disorders	Preclinical	Rodent self-administration, conditioned place preference	Rats, mice, zebra fish	Ayahuasca (various doses)	5-HT _{2A} antagonist (in some studies)	↓ Ethanol intake, ↓ Amphetamine self-administration, ↓ conditioned place preference	Behavioral paradigms, Fos expression
	Clinical	Observational, surveys, longitudinal naturalistic studies	Church members, retreat participants	Ritual ayahuasca use	None (observational)	↓ Alcohol, tobacco, cannabis use; ↑ Well-being	AUDIT, DASS-21, PANAS
Depression	Preclinical	Forced swim test, chronic stress models	Rats, mice, marmosets	Ayahuasca or harmine	Antidepressants (fluoxetine) or vehicle	↓ Immobility, ↑ BDNF, levels, ↓ cortisol levels	Forced swim test, BDNF and levels
	Clinical	Open-label, RCTs	TRD patients ($n=6-29$)	Single ayahuasca dose, DMT	Placebo (in RCT)	↓ depressive symptoms, ↓ suicidality	HAM-D, MADRS, MADRS-SI
PTSD	Preclinical	Fear conditioning, extinction	Rats	Ayahuasca, DMT	Vehicle	↓ Freezing, ↑ Fear extinction retention	Freezing behavior
	Clinical	Observational, retreat studies	Trauma survivors, veterans	Ayahuasca ceremonies (1–3 sessions)	None (observational)	↓ PTSD symptoms, ↑ resilience, ↑ purpose	CAPS-5, PCL-5
Anxiety	Preclinical	Elevated plus maze, open field, zebrafish assays	Rodents, zebrafish	Ayahuasca (low vs. high doses)	Vehicle	Dose-dependent anxiolytic vs. anxiogenic	EPM, locomotion
	Clinical	Observational, RCTs	Ritual users, naïve volunteers	Single or repeated ritual use	Placebo (in RCT)	↓ GAD symptoms, ↓ panic-like symptoms, ↑ self-perception	STAI, BAI, PANAS
Eating Disorders	Preclinical	—	—	—	—	—	—
	Clinical	Observational, cross-sectional	Individuals with ED history	Ayahuasca rituals	None	↑ Body image, ↓ compulsive eating, ↑ well-being	Self-reports, EDE-Q
Personality Disorders	Preclinical	—	—	—	—	—	—
	Clinical	Observational, longitudinal, RCT (subset of TRD with PD comorbidity)	Long-term users, TRD patients	Ritual or single-dose Ayahuasca	Placebo (in RCT)	↓ Neuroticism, ↑ Openness, ↑ Emotion regulation	NEO-PI-R, mindfulness scales, MADRS

AUDIT Alcohol Use Disorder Identification Test, *DASS-21* Depression, Anxiety and Stress Scale, *PANAS* Positive and Negative Affect Scale, *BDNF* Brain derived neurotrophic factor, *RCT* Randomized Clinical Trial, *TRD* Treatment resistant depression, *DMT* N,N-dimethyltryptamine, *HAM-D* Hamilton Depression Rating Scale, *MADRS* Montgomery-Åsberg Depression Rating Scale, *MADRS-SI* Suicidal ideation item of MADRS, *PTSD* Post-traumatic stress disorder, *CAPS-5* Clinician-Administered PTSD Scale for DSM-5, *PCL-5* PTSD Checklist for DSM-5, *EPM* Elevated plus maze, *GAD* Generalized Anxiety Disorder, *STAI* State Anxiety Inventory-State Scale, *BAI* Beck Anxiety Inventory, *ED* Eating Disorders, *EDE-Q* Eating Disorder Examination Questionnaire, *PD* Personality disorder, *NEO-PI-R* Revised NEO Personality Inventory

study of UDV members, found no organ toxicity or cognitive impairment. On the contrary, participants demonstrated improved psychological well-being and lower rates of substance abuse [12, 13]. Adolescent participants also showed no adverse effects [15, 16].

A large-scale global survey confirmed vomiting (69.9%) as the most common side effect, followed by transient psychological discomfort, typically mild in controlled environments [17, 18]. Preclinical studies corroborate these findings, showing that low to moderate doses are well tolerated in animal models; higher doses may induce more significant side effects, while overdose occurs with 50 times the doses used in ceremonial context [19, 20]. Although human data remain scarce, animal studies related to pregnancy have found that high doses can affect fetal development [21]. At typical doses, even repeated use did not result in organ damage [20, 22], though long-term high-dose exposure raised concerns about weight gain and neurotoxicity [23, 24].

Some psychological risks remain, particularly in individuals with pre-existing psychiatric conditions. Ayahuasca may exacerbate symptoms of bipolar disorder or schizophrenia and induce anxiety or dysphoria in unprepared individuals [25]. Theoretically, interactions with serotonergic drugs like SSRIs can increase the risk of serotonin syndrome [26, 27], emphasizing the need for careful supervision.

The psychological effects of ayahuasca are also modulated by extrapharmacological factors, such as the user's mindset, expectations, and environment (set and setting), which can influence therapeutic outcomes and risk of adverse reactions. These effects highlight the importance of implementing appropriate screening, preparation, and integration practices in both ceremonial and clinical settings [28, 29].

Substance Use Disorders

Preclinical Evidence

Preclinical evidence from animal models has suggested potential therapeutic effects of ayahuasca and its alkaloids in SUD [30]. A systematic review encompassing 32 studies across rodents, primates, and zebrafish reported that ayahuasca reduced behavioral markers of drug reward, including ethanol and amphetamine self-administration, while showing a favorable safety profile at ceremonial-equivalent doses [20]. For instance, ayahuasca has been shown to attenuate ethanol-induced place preference and behavioral sensitization in mice significantly [31, 32] and reduce voluntary ethanol intake in rats [31]. These effects were often reversed by 5-HT_{2A} antagonists, underscoring the role of serotonergic mechanisms [32]. Furthermore, ayahuasca modulated ethanol-induced neuroplasticity and protein

expression in brain regions such as the hippocampus and striatum [33]. It reversed methylphenidate-induced conditioned place preference while normalizing associated Fos expression patterns [34].

Adding to this evidence, a recent study using a two-bottle choice paradigm demonstrated that male mice voluntarily prefer water over ayahuasca depending on dose and exposure schedule [35]. Mice exposed to low concentrations of ayahuasca (0.01–0.03 mg/mL) every other day showed a preference for the brew, while those exposed less frequently or to higher concentrations developed aversion. These findings suggest that both frequency of use and dosage are critical variables that modulate the rewarding versus aversive effects of ayahuasca. The authors propose that serotonergic receptor occupancy, particularly differential engagement of 5-HT_{2A} versus 5-HT_{2C} receptors, may underlie these dose-dependent motivational effects [32].

Another review reinforced these findings, reporting that ayahuasca reduced amphetamine self-administration and conditioned place preference for alcohol and methylphenidate in rodents [36]. These effects appear to be mediated by the synergistic action of MAO-A inhibitors and DMT as a 5-HT_{2A} receptor agonist [37].

Clinical Evidence

Observational studies have consistently reported reductions in substance use among long-term ritual ayahuasca users. Online surveys further confirm that ayahuasca users, including those with previous SUD, report less current substance use and greater well-being compared to other psychedelic or nonpsychedelic drug users [38]. The Global Ayahuasca Survey also found that individuals with prior substance use concerns reported sustained reductions following ritual use of the brew, alongside improvements in psychological well-being [39].

Early evidence already suggested the potential positive benefit of ayahuasca in alcohol addiction [12]. One large-scale study found significantly lower rates of current alcohol and tobacco use disorder among members of UDV compared to national Brazilian norms, with more potent effects observed in those with more extended membership and more frequent ceremony attendance [40]. In addition, a naturalistic longitudinal study of first-time ayahuasca participants found reductions in alcohol and cannabis use, depression, anxiety, and stress one month after the ceremony, along with increases in self-efficacy and psychological flexibility [41]. A comprehensive narrative review highlighted that participants in traditional Amazonian addiction treatment centers reported reduced drug use and craving after ayahuasca-assisted therapy. At the same time, Western retreat and church settings have shown associations between ayahuasca

use and lower alcohol, tobacco, and other drug consumption across both adolescent and adult populations [42–46].

Despite promising observational data, the lack of RCTs remains a key limitation. Most evidence relies on retrospective reports and survey data, making causal inference difficult. Moreover, variations in dose, setting, and participant characteristics across studies hinder the generalization of findings. Nevertheless, the consistency of findings across multiple cohorts and contexts suggests that ayahuasca may offer a novel therapeutic approach for SUD, particularly when embedded within structured and supportive environments [47].

Depression

Preclinical Evidence

Preclinical studies primarily utilize rodent models of depression, such as the forced swim test (FST), chronic unpredictable stress (CUS) paradigm, and learned helplessness models, to assess ayahuasca's behavioral and neurobiological antidepressant effects. These models consistently suggest that ayahuasca reduces immobility time in the FST, a widely used marker of antidepressant efficacy [20].

Several studies have demonstrated that a single ayahuasca administration produces rapid antidepressant effects. In male rats, ayahuasca increased exploratory behavior and hyperkinesia within 10 min of administration, with effects normalizing within an hour [48]. Furthermore, in the FST, both low and high doses of ayahuasca (2.5–10 mg/kg of DMT) significantly decreased immobility time, suggesting an antidepressant-like response [49]. A similar effect was observed in female rats, where ayahuasca increased swimming activity and reduced immobility more effectively than fluoxetine [22].

Harmin has shown promising outcomes. Acute (5–15 mg/kg) and chronic (14-day) administration reduced immobility in the FST and increased hippocampal brain-derived neurotrophic factor (BDNF) levels in rats [50–52]. Harmin also modulated mitochondrial function and oxidative stress markers in the prefrontal cortex and hippocampus, suggesting potential pathways through which it may exert its antidepressant effects [53].

Animal models also suggest that ayahuasca may have preventive effects. A study in juvenile marmosets subjected to chronic social isolation found that a single ayahuasca administration reduced both behavioral and physiological markers of depressive-like states [54]. Moreover, repeated administrations were found to prevent stress-induced anhedonia and maintain normal cortisol responses. In contrast, untreated marmosets exhibited depressive-like behaviors and blunted stress reactivity; those receiving ayahuasca

retained behavioral and endocrine profiles similar to non-isolated controls [55]. Its prophylactic action also mitigated neuronal volume loss in striatal neurons, which is typically induced by prolonged social isolation [56]. These findings suggest that ayahuasca may modulate neuroplasticity, promote resilience to chronic stress, and prevent the onset of depressive symptoms [55, 56].

Preclinical research has demonstrated that ayahuasca exerts significant neurobiological effects that may underlie its antidepressant properties. Studies in animal models have shown that ayahuasca enhances synaptic plasticity, promotes neurogenesis, and modulates key neurotransmitter systems involved in mood regulation. Specifically, it increases serotonin receptor activity (5-HT_{2A} and 5-HT_{2C}), upregulates BDNF, and influences glutamatergic neurotransmission [19]. These mechanisms contribute to long-lasting neuroplastic changes, which may explain the sustained antidepressant effects observed after a single administration [20]. Furthermore, ayahuasca's MAO-A inhibition potentially prolongs serotonin availability and enhances mood-stabilizing impact [57]. Ayahuasca also seems to modulate the reward system, influencing dopamine-related pathways in a manner that distinguishes it from classic antidepressants [58]. While the serotonergic system plays a central role, dopaminergic and glutamatergic mechanisms mediate its sustained effects, reinforcing the hypothesis that ayahuasca induces neurochemical adaptations relevant to depression treatment [20].

Beyond neurotransmitter modulation, ayahuasca has been found to mitigate stress-induced neuroinflammation and oxidative damage in brain regions implicated in depression, including the hippocampus and prefrontal cortex [59]. Chronic administration studies report increased c-Fos expression, an indicator of neuronal activation, in the amygdala, prefrontal cortex, and hippocampus, suggesting enhanced connectivity and neural plasticity [20]. These findings align with research on other rapid-acting antidepressants, such as ketamine, which also induces dendritic remodeling and activates the mammalian target of rapamycin (mTOR) pathway, a critical regulator of synaptic plasticity [60].

Clinical Evidence

Initial reports on the antidepressant potential of ayahuasca in humans came from participants in traditional ceremonies [61, 62]. Early studies on members of syncretic churches suggested that regular ayahuasca use was not linked to psychopathological, cognitive, or personality deterioration, but rather to potential mental health benefits [12, 63, 64]. Online surveys supported these observations, indicating that many users experienced significant alleviation of depressive

symptoms. An international survey involving over 11,000 participants suggested that ayahuasca use was associated with substantial antidepressant effects [65]. Significant reductions in depressive symptoms were also reported by participants in ayahuasca retreats with improvements sustained for up to six months after the retreat [41, 66, 67]. Furthermore, evidence indicates that naturalistic use of ayahuasca is linked to improved mental health outcomes and greater psychological well-being, both in individuals with a history of mental illness and those without such diagnoses [39].

Building on these observational findings, three clinical trials have assessed ayahuasca's antidepressant effects. The first open-label trial with six patients suffering from treatment-resistant depression found significant symptom reductions within 24 h, lasting up to 21 days [68]. A subsequent open-label study with 17 patients showed similar results, with improvements observed as early as 40 min post-ingestion and persisting for 21 days [69]. A more robust randomized placebo-controlled trial with 29 patients confirms significant antidepressant effects of ayahuasca compared to placebo, with symptom reductions lasting at least a week [70], which was accompanied by reduced suicidality from baseline to several days post-session [71].

Also, in recent years, the therapeutic potential of DMT, when administered in purified form, independent of the ayahuasca brew, has garnered increasing scientific attention. Preclinical, neuroimaging, and early-phase clinical studies suggest that DMT is safe and well-tolerated when used in controlled settings, with promising rapid-acting antidepressant [72–74].

Controlled human studies have begun to systematically explore the effects of isolated DMT. A placebo-controlled study that combined variable intravenous doses (7–20 mg) with a fixed-dose arm (20 mg) in control subjects [75]. The results revealed reductions in both depression and anxiety scores lasting up to two weeks post-administration in controls. Notably, these improvements were positively correlated with the intensity of the psychedelic experience. Also, an exploratory open-label, fixed-order, dose-escalation phase 1 study investigated the safety, tolerability, and preliminary efficacy of intravenous DMT in treatment-resistant major depressive disorder (MDD) and healthy controls. Ten participants (7 MDD, 3 healthy) received 0.1 mg/kg followed by 0.3 mg/kg DMT. DMT was generally well tolerated, with transient increases in blood pressure, heart rate, and anxiety resolving within 30 min. Depression severity (HAMD-17) significantly decreased one day after the 0.3 mg/kg dose. Adverse events were mostly mild, and no participants discontinued [76]. Taken together, these data suggest that DMT's ultra-short duration of action may enable

flexible and potentially repeatable dosing strategies in future therapeutic settings.

In parallel, the effects of vaporized DMT have also been assessed in clinical trials. A Phase I study involving 27 healthy participants established a favorable safety and tolerability profile, characterized by elevated positive affect and mild, transient physiological changes [73]. Building on these findings, a subsequent open-label Phase IIa trial enrolled six patients with TRD. Following a single administration of inhaled DMT, the mean MADRS score decreased by 22 points after seven days and by 17 points at the one-month follow-up. At the one-week mark, 83% of participants were classified as responders, with 67% in remission. After one month, 67% of the participants retained their response, and half remained in remission [74]. In a subsequent open-label trial, 14 patients with TRD received inhaled DMT in a dose-escalation design (15 mg and 60 mg) [77]. The intervention was safe and well-tolerated, producing manageable psychedelic effects and no serious adverse events, including among participants using concurrent antidepressant medications. Rapid antidepressant effects were observed. Clinical response rates were 79% at day 1, rising to 86% at both days 7 and 14. At one month, 64% still met response criteria, and 57% at three months. Remission rates followed a similar pattern: 71% at D1, 57% at D7 and D14, 57% at one month, and 36% at three months. Suicidal ideation was significantly reduced. At baseline, 86% of participants reported some level of suicidality, including 43% with severe ideation. By the day after dosing, no patients had severe suicidal thoughts, and 79% reported no suicidal ideation. This effect remained consistent, with only 21% of participants reporting any suicidality at three months [77].

Neuroimaging studies further illuminate DMT's mechanisms of action. Functional MRI data suggest that DMT temporarily disrupts the integrity of the default mode network (DMN) while simultaneously enhancing global functional integration, a neural change often associated with reductions in depressive symptoms and increased emotional adaptability [78].

At the molecular level, preclinical and translational research support DMT's ability to enhance neuroplasticity and neurogenesis, potentially through the upregulation of BDNF and downstream signaling pathways involved in synaptic remodeling and stress resilience [79, 80]. These neurobiological effects may underlie the sustained clinical improvements observed in depression and anxiety outcomes.

Post-traumatic Stress Disorder (PTSD)

Although research on ayahuasca's therapeutic effects on PTSD remains limited, emerging evidence suggests it may facilitate trauma processing and emotional regulation [47].

Animal and in vitro studies suggest that ayahuasca may modulate conditioned fear responses and influence PTSD-related symptoms [81], and in humans, evidence is limited to observational studies [44].

Preclinical Evidence

In fear conditioning models, ayahuasca or isolated DMT administration suppressed freezing behavior, indicating potential modulation of fear responses [81]. Additionally, repeated administration of moderate doses of ayahuasca infusion promoted the retention of fear extinction in Wistar rats, suggesting that the substance may contribute to the restructuring of traumatic memories [82].

Beyond facilitating fear extinction, ayahuasca has also demonstrated the ability to interfere with the reconsolidation of traumatic memories. In a rodent study, oral administration of ayahuasca immediately after the reactivation of an aversive memory resulted in a sustained reduction in freezing behavior in the following days, indicating disruption of traumatic memory restabilization [82]. These results suggest that ayahuasca may act as a modulator of emotional responses associated with traumatic memories, potentially weakening the intensity of aversive recollections.

Clinical Evidence

Qualitative research has indicated that individuals self-identifying as having PTSD often describe processing past traumas during ayahuasca-induced visions. This experience is associated with improved trauma-related symptoms, emotional well-being, and overall functional outcomes [83]. A longitudinal observational study with 63 self-selected participants with histories of trauma examined the effects of ayahuasca in a naturalistic ceremonial setting. Results indicated sustained symptom relief, particularly in individuals with severe pre-treatment PTSD scores [66]. A recent mixed-methods study investigated the effects of a three-day ayahuasca retreat on military veterans with PTSD symptoms. By post-treatment, 71.4% of participants exhibited clinically meaningful reductions in PTSD symptoms, with these improvements sustained at a three-month follow-up. Participants also reported increased emotional resilience, acceptance, and a renewed sense of purpose, though some experienced acute psychological distress during the ceremonies [84].

Anxiety

Preclinical Evidence

Preclinical studies in animal models of anxiety have yielded mixed but informative results. A systematic review analyzed

32 studies using rodents, zebrafish, and primates, highlighting that while the brew demonstrates safety at doses comparable to those used in ceremonies, findings regarding its anxiolytic potential are inconclusive [20]. Some studies using the elevated plus maze (EPM) and open field tests suggest reduced anxiety-like behavior following ayahuasca or its alkaloids administration. Still, other reports showed no effect or even anxiogenic effects, depending on the dose, duration, and species [20].

In zebrafish models, low doses of ayahuasca (0.1 ml/L) acutely reduced anxiety-like behaviors, as indicated by decreased bottom-dwelling and freezing. Higher doses (1.0–3.0 ml/L) led to pronounced anxiogenic responses and reduced locomotion [85]. In rodents, ayahuasca prevented and reversed anxiety-like behavior induced by lipopolysaccharide-mediated neuroinflammation in rats using the open field and FST [86]. These effects were observed within 48 h, suggesting a rapid anxiolytic and antidepressant action. Neurobiological markers of reduced inflammation and oxidative stress accompanied behavioral improvements.

On the other hand, higher doses of ayahuasca (30–50× the ceremonial dose) produced clear signs of toxicity and anxiogenic-like behavior in female rats [22]. These included tremors, piloerection, and significantly reduced exploration in the EPM, as well as reduced locomotion and increased immobility, pointing to heightened anxiety and motor suppression. Notably, these effects were not observed at lower doses, emphasizing the dose-dependent nature of ayahuasca's behavioral outcomes [22].

The alkaloids harmine and harmaline have also been studied in isolation. In one study, harmaline showed a biphasic effect in mice: low doses (5–10 mg/kg) increased anxiety, while higher doses (20 mg/kg) exerted anxiolytic effects in the EPM [87]. A study in a rat model of sepsis-induced neuroinflammation suggests that pretreatment with ayahuasca (for 3 days) prevented anxiety-like behaviors in both the EPM and open field tests. These effects were accompanied by increased levels of IL-4 and BDNF, as well as reduced oxidative stress markers, in brain regions associated with mood regulation [88].

Clinical Evidence

In human studies, early evidence of anxiolytic effects came from observational and anecdotal reports among ritual ayahuasca users. These individuals often report reduced anxiety symptoms and enhanced emotional regulation [44, 47]. The Global Ayahuasca Project, an extensive international survey ($n = 11,912$), found that 70% of participants who reported pre-ceremony anxiety experienced improvement, and 16% reported complete resolution of symptoms [65]. Despite these promising outcomes, methodological limitations must

be acknowledged, such as a lack of controls and self-selection bias.

Importantly, preclinical evidence aligns with observational human studies. Field studies conducted in naturalistic and religious contexts support these findings. For instance, a review of seven studies focused on GAD symptoms noted consistent reductions in post-ayahuasca consumption [89]. A double-blind, placebo-controlled design to assess the acute effects of ayahuasca in long-term Santo Daime users found that ayahuasca significantly reduced scores of panic-like symptoms and hopelessness. However, it did not alter state or trait anxiety levels, possibly due to low baseline anxiety in the experienced sample [64]. A double-blind, randomized trial with ayahuasca-naïve participants ($n = 17$) found that one dose (2 ml/kg) improved self-perception but not anxiety in a public speaking test [90].

Taken together, evidence from animal models, observational studies, and preliminary clinical trials suggests that ayahuasca may have anxiolytic effects, particularly in contexts involving neuroinflammation or ritual use. However, outcomes are dose-dependent, and high doses may be anxiogenic.

Eating Disorders (ED)

Clinical Evidence Ayahuasca has emerged as a potential therapeutic tool, offering deep introspection and emotional processing, which can be particularly beneficial for individuals with ED [91–93]. Participants in ayahuasca ceremonies report profound shifts in self-perception, allowing them to address the underlying emotional and psychological factors contributing to their disordered eating [94]. By facilitating the reevaluation of self-worth and body image, ayahuasca has demonstrated promise in helping individuals move toward healing and recovery [95].

Preliminary studies suggest that ayahuasca may offer therapeutic benefits for individuals with ED by enhancing emotional processing and promoting psychological insight [93, 96]. A cross-sectional study examining individuals with a history of ED who participated in ayahuasca ceremonies and found significant self-reported improvements in eating behaviors, body image, and overall well-being [91]. These findings align with broader research on psychedelics, which suggests that heightened interconnectivity in brain networks plays a role in reducing compulsive behaviors [97]. Given the high prevalence of trauma in individuals with ED, ayahuasca's capacity to modulate fear responses and facilitate emotional breakthroughs may also be relevant in addressing underlying psychological factors that contribute to disordered eating, similar to the effects observed with other psychedelics [98].

The healing potential of ayahuasca in ED is thought to stem from its capacity to elicit powerful personal insights and foster emotional resilience. Many participants describe their experiences as transformative, helping them confront deep-seated trauma and negative self-beliefs that fuel disordered eating behaviors [99]. A study involving ceremony leaders found that ayahuasca supports a holistic healing process by promoting a sense of interconnectedness and self-compassion, which are critical for ED recovery [96]. However, concerns have been raised regarding some aspects of the ayahuasca ritual, such as the required dietary restrictions and purging, which might reinforce disordered behaviors in vulnerable individuals [91]. Nonetheless, many participants report that these aspects of the ceremony, when framed within the context of healing and spiritual cleansing, help recontextualize their relationship with food and their bodies [94].

Personality Disorders

Clinical Evidence Ayahuasca has been investigated for its potential to modulate personality traits, particularly those related to emotional dysregulation and interpersonal difficulties, which are hallmark features of personality disorders [100, 101]. An observational study found that ayahuasca significantly improved emotion regulation and mindfulness capacities in a group of volunteers, including individuals with borderline-like traits. Participants showed improved emotional acceptance, reduced emotional interference, and better emotional control, suggesting that ayahuasca may modulate core mechanisms underlying emotion regulation [102, 103]. Similarly, an RCT involving individuals with treatment-resistant depression, several of whom had comorbid Cluster B personality disorders, found that a single dose of ayahuasca resulted in significant reductions in depressive symptoms and suicidality [70, 71].

Long-term use of ayahuasca has been linked to structural brain changes, which may underlie its effects on personality traits, as in a study that investigated the impact of chronic ayahuasca use on cortical thickness in regular users compared to matched controls [104]. The results showed significant cortical thinning in midline brain structures, particularly the posterior cingulate cortex (PCC), a key hub of the default mode network (DMN). Interestingly, PCC thinning was inversely correlated with self-transcendence, a personality trait associated with spirituality, transpersonal feelings, and a reduced sense of self-centeredness.

In addition, studies have reported shifts in personality traits following ayahuasca use. Research suggests that these substances can increase openness to experience, a trait associated with cognitive flexibility and receptivity to new experiences

[100, 101]. However, the effects of ayahuasca on personality remain inconsistent. A recent study found that while some participants exhibited increased openness three weeks after ayahuasca ingestion, others did not show significant changes, suggesting that individual differences and methodological factors may influence the effects [101]. Additionally, individuals who participated in traditional ayahuasca ceremonies demonstrated sustained reductions in neuroticism and increases in agreeableness up to six months post-ingestion. These changes were particularly pronounced among participants who reported more intense mystical experiences [105].

Ayahuasca has also been associated with robust changes in core personality dimensions. In a prospective longitudinal study, significant reductions in Neuroticism and increases in Extraversion, Openness, Agreeableness, and Conscientiousness were observed following ceremonial ayahuasca use [106]. These changes were evident in self-report and informant-report measures. They were moderated by baseline personality traits and the intensity of mystical-type and reappraisal experiences during the ceremony.

The available evidence suggests that ayahuasca may have a role in improving emotion regulation and modifying personality traits of long-term users [107], which could offer potential therapeutic benefits, particularly for those individuals characterized by emotional instability and maladaptive interpersonal patterns [107]. Nonetheless, some caution is warranted. While historical and contemporary case reports highlight meaningful improvements in traits such as narcissism and psychopathy, adverse reactions, such as acute anxiety or dissociation, have also been observed, particularly in patients with emotional instability or comorbidities [108]. This reinforces the need for careful screening and integration protocols when considering psychedelic therapy for personality disorders.

General Discussion

Ayahuasca has emerged as a promising candidate in psychedelic therapy, with a growing body of evidence supporting its potential therapeutic effects across a range of mental health conditions. Importantly, this broad applicability does not necessarily indicate that ayahuasca functions as a universal remedy. Instead, it appears to engage transdiagnostic mechanisms, such as emotional regulation, cognitive flexibility, and trauma processing, that cut across conventional psychiatric classifications [41, 109].

These therapeutic effects result from a synergistic interplay of neurobiological and psychological processes. Pharmacologically, ayahuasca's active compounds, DMT and β -carbolines, interact with serotonin receptors (notably 5-HT_{2A}), inhibit MAO-A, and modulate dopaminergic and glutamatergic signaling. These actions are associated with

enhanced neuroplasticity, reduced inflammation, and alterations in stress-related pathways [8, 58, 110–112]. Additionally, human neuroimaging studies reveal altered activity and connectivity in brain networks linked to emotional regulation and self-perception, particularly the DMN and the salience network [107, 113, 114].

On the subjective level, the ayahuasca experience frequently involves intense visual imagery, emotional breakthroughs, and mystical-type states [7, 115]. These psychological phenomena have been consistently linked to therapeutic outcomes [70, 116, 117]. For instance, greater intensity of perceptual and emotional experiences has been associated with more substantial reductions in depressive symptoms [70]. Such experiences may facilitate reprocessing trauma, resolving inner conflict, and cultivating self-compassion, which are critical for recovery from chronic and treatment-resistant conditions [118, 119].

Neurophysiological findings further reinforce this model. Ayahuasca has been shown to increase brain entropy and the redundancy of functional connectivity networks, markers of cognitive flexibility and psychological resilience [120, 121]. Biomarker studies reveal consistent trends, including normalization of cortisol levels [8], elevation of serum BDNF [110], and reduction in systemic inflammation, as measured by C-reactive protein [111]. In PTSD specifically, a compelling hypothesis posits that ayahuasca exerts its therapeutic effects via epigenetic modulation of the sigma-1 receptor, which is involved in neuroplasticity, stress response, and memory integration. Post-ayahuasca increases in sigma-1 receptor methylation have been observed in individuals with a history of childhood trauma, suggesting a potential molecular pathway for the reprocessing of suppressed traumatic memories [66, 122].

Despite this encouraging picture, several limitations must be acknowledged. Most human studies to date are observational in nature and often lack placebo control conditions or blinding procedures. While some RCTs have been conducted, particularly in depression, the number remains relatively small [70, 71]. One key methodological challenge in psychedelic research concerns expectancy effects. Participants' beliefs, motivations, and prior experiences can strongly influence outcomes. These expectancy effects may amplify perceived improvements, especially in open-label or non-blinded settings. Relatedly, the issue of achieving a trustworthy placebo in psychedelic trials presents a significant limitation, due to the unmistakable subjective effects of psychedelics. Blinding participants, and often researchers, is inherently tricky. In addition, safety considerations remain paramount. Individuals with a history of psychotic disorders, bipolar disorder, or severe personality disorders may be at elevated risk for adverse psychological reactions [25, 123]. Finally, the ethical challenges posed by the global

expansion of ayahuasca use must be carefully considered. These include issues such as cultural appropriation, ecological sustainability, and the commodification of Indigenous knowledge systems. The integration of ayahuasca into contemporary therapeutic contexts requires not only scientific rigor but also respectful and reciprocal engagement with the Indigenous communities who have preserved and cultivated this medicinal tradition for generations [124].

Conclusions

Ayahuasca represents a novel and multifaceted intervention that operates across neurobiological, psychological, and spiritual dimensions. Its potential to target transdiagnostic processes offers a unique contribution to mental health care, particularly for individuals unresponsive to conventional treatments. Future research, grounded in rigorous methodology and cultural humility, will be crucial for further elucidating its mechanisms, refining its applications, and ensuring its ethical and sustainable use. When thoughtfully integrated, ayahuasca can expand our therapeutic toolkit and deepen our understanding of consciousness, healing, and the human condition.

Key References

- White E, Kennedy T, Ruffell S, Perkins D, Sarris J. Ayahuasca and dimethyltryptamine adverse events and toxicity analysis: A systematic thematic review. *Int J Toxicol*. 2024;43:327–39.
 - This review assesses the safety and toxicity of ayahuasca and its alkaloids, supporting a favorable safety profile in controlled settings.
- Daldegan-Bueno D, Simionato NM, Favaro VM, Maia LO. The current state of ayahuasca research in animal models: A systematic review. *Prog Neuropsychopharmacol Biol Psychiatry*. 2023;125:110738.
 - This review summarizes findings from animal studies on ayahuasca, suggesting safety at ceremonial doses and therapeutic potential for depression and substance use, though anxiolytic effects remain unclear.
- Marinho EAV, Serra YA, Oliveira-Lima AJ, Marcourakis T, Berro LF. Ayahuasca for the treatment of alcohol use disorder. *Int Rev Neurobiol*. 2024;178:283–300.
 - This chapter reviews clinical and preclinical evidence supporting ayahuasca's potential as a treatment for alcohol use disorder, highlighting its impact on reducing alcohol consumption and related behaviors.
- Perkins D, Sarris J, Cowley-Court T, Aicher H, Tófoli LF, Carlos Bouso J, et al. Associations between ayahuasca use in naturalistic settings and mental health and wellbeing outcomes: Analysis of a large global dataset. *J Psychoactive Drugs*. 2024;1–12.
 - This study links naturalistic ayahuasca use to improved mental health and well-being, moderated by experiential and integration-related factors.
- Maia LO, Daldegan-Bueno D, Wießner I, Araujo DB, Tófoli LF. Ayahuasca's therapeutic potential: What we know - and what not. *Eur Neuropsychopharmacol*. 2023;66:45–61.
 - This review summarizes evidence on ayahuasca's therapeutic effects across various physical and mental health conditions.
- Rush B, Marcus O, García S, Loizaga-Velder A, Spitalier A, Mendeive F. Ayahuasca treatment outcome project (ATOP): One-year results from takiwasi center and implications for psychedelic science. *J Stud Alcohol Drugs*. 2024;85:607–18.
 - This study reports positive 1-year outcomes from an ayahuasca-assisted addiction treatment program, suggesting potential benefits within a multifactorial therapeutic context.
- Sheth R, Parikh E, Olayeye K, Pfeifer K, Khanna D. The effects of ayahuasca on psychological disorders: A systematic literature review. *Cureus*. 2024;16:e55574.
 - This systematic review highlights ayahuasca's therapeutic potential across various mental health conditions, emphasizing the role of the mystical experience and ritual context in its effects.
- de Lira-Bandeira WG, de Mendonça Batista LAC, de Medeiros Bandeira AS, de Góis Moraes PLA, Pereira LRF, de Meiroz Grilo MLP, et al. Ayahuasca partially preserves striatal integrity in juvenile non-human primates exposed to chronic stress: evidence from stereological evaluation. *Front Neuroanat*. 2025;19:1457557.

- This study in a primate model of juvenile depression suggests that ayahuasca may prevent striatal neuronal volume loss caused by chronic social isolation, indicating potential neuroprotective effects.
- Weiss B, Dinh-Williams L-AL, Beller N, Raugh IM, Strauss GP, Campbell WK. Ayahuasca in the treatment of posttraumatic stress disorder: Mixed-methods case series evaluation in military combat veterans. *Psychol Trauma*. 16(Suppl 3), S718–S722.
 - This mixed-methods case series suggests that a brief ayahuasca intervention may produce lasting reductions in PTSD and mood symptoms among military veterans.
- de Camargo RW, Joaquim L, Machado RS, de Souza Ramos S, da Rosa LR, de Novais Junior LR, et al. Ayahuasca pretreatment prevents sepsis-induced anxiety-like behavior, neuroinflammation, and oxidative stress, and increases brain-derived neurotrophic factor. *Mol Neurobiol*. 2025;62(5):5695–5719.
 - Ayahuasca prevented anxiety-like behaviors, neuroinflammation, and oxidative stress in septic rats, suggesting neuroprotective and anxiolytic effects.
- Arruda Sanchez T, Ramos LR, Araujo F, Schenberg EE, Yonamine M, Lobo I, et al. Emotion regulation effects of Ayahuasca in experienced subjects during implicit aversive stimulation: An fMRI study. *J Ethnopharmacol*. 2024;320:117430.
 - This study investigates Ayahuasca's acute emotional effects on brain response to aversive stimuli using fMRI, revealing that Ayahuasca attenuates amygdala activity and enhances insular and prefrontal cortex activation, alongside reduced anxiety and mental sedation.
- Omágua-Kambebe A, Labate BC, Ribeiro S. Psychedelic science and Indigenous shamanism: an urgent dialogue. *Nat Ment Health*. 2023;1:815–6.
 - The global rise of psychedelic therapies and ayahuasca highlights the importance of Indigenous knowledge and rituals in healing, emphasizing cultural respect and holistic approaches in therapy.

Author Contributions The first draft of the manuscript was written by D.A. and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). Studies from our group cited in this review were funded by the CAPES Foundation, the Brazilian National Council for Scientific and Technological Development (CNPq), and the Financier of Studies and Projects (FINEP). The Federal University of Rio Grande provided institutional support do Norte (UFRN) and Onofre Lopes University Hospital (HUOL).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Human and Animal Rights and Informed Consent All reported studies/experiments with human or animal subjects performed by the authors have been previously published and complied with all applicable ethical standards (including the Helsinki declaration and its amendments, institutional/national research committee standards, and international/national/institutional guidelines).

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Schultes RE, Hofmann A, Rätsch C. Plants of the gods: their sacred, healing, and hallucinogenic powers. Rev. and e. Rochester, Vt.: Healing Arts Press; 2001.
2. Miller MJ, Albarracin-Jordan J, Moore C, Capriles JM. Chemical evidence for the use of multiple psychotropic plants in a 1,000-year-old ritual bundle from South America. *Proc Natl Acad Sci U S A*. 2019;116:11207–12.
3. Labate BC, Jungaberle H. The internationalization of ayahuasca. Zürich.: LIT Verlag Münster; 2011. p. 446.
4. Rivier L, Lindgren JE. Ayahuasca, South-American hallucinogenic drink - ethnobotanical and chemical investigation. *Econ Bot*. 1972;26:101–29.
5. McKenna DJ. Ayahuasca: an ethnopharmacologic history. *Sacred Vine Spirits: Ayahuasca*. 1998;1:1–50.
6. Riba J, Rodriguez-Fornells A, Urbano G, Morte A, Antonijoan R, Montero M, et al. Subjective effects and tolerability of the South American psychoactive beverage Ayahuasca in healthy volunteers. *Psychopharmacology*. 2001;154:85–95.
7. Shanon B. The antipodes of the mind: charting the phenomenology of the Ayahuasca experience. In: <book-title update="added">The antipodes Of the mind: Charting the Phenomenology of the Ayahuasca Experience. 1st ed. Oxford: Oxford University Press; 2003. p. 496.

8. Galvão AC, de Almeida M, Silva RN, Freire EADS, Palhano-Fontes FAM, Onias F. Cortisol modulation by Ayahuasca in patients with treatment resistant depression and healthy controls. *Front Psychiatry*. 2018;9:185.
9. Shanon B. The epistemics of Ayahuasca visions. *Phenomenol Cogn Sci*. 2010;9:263–80.
10. Spruce R. Notes of a Botanist on the Amazon & Andes: Being Records of Travel on the Amazon and Its Tributaries, the Trombetas, Rio Negro, Uaupés, Casiquiari, Pacimoni, Huallaga and Pastaza; as Also to the Cataracts of the Orinoco, Along the Eastern Side of the Andes of Peru and Ecuador, and the Shores of the Pacific, During the Years 1849–1864. Macmillan; 1908.
11. Labate BC, Feeney K. Ayahuasca and the process of regulation in Brazil and internationally: implications and challenges. *Int J Drug Policy*. 2012;23:154–61.
12. Grob CS, McKenna DJ, Callaway JC, Brito GS, Neves ES, Oberlaender G, et al. Human psychopharmacology of hoasca, a plant hallucinogen used in ritual context in Brazil. *J Nerv Ment Dis*. 1996;184:86–94.
13. McKenna DJ, Callaway JC, Grob CS. The scientific investigation of ayahuasca: a review of past and current research. *Heffter Rev Psyched Res*. 1998;1:195–223.
14. Guimarães dos Santos R. Safety and side effects of Ayahuasca in humans—an overview focusing on developmental toxicology. *J Psychoact Drugs*. 2013;45:68–78.
15. Da Silveira DX, Grob CS, de Rios MD, Lopez E, Alonso LK, Tacla C, et al. Ayahuasca in adolescence: a preliminary psychiatric assessment. *J Psychoactive Drugs*. 2005;37:129–33.
16. de Rios MD, Grob CS, Lopez E, da Silveira DX, Alonso LK, Doering-Silveira E. Ayahuasca in adolescence: qualitative results. *J Psychoactive Drugs*. 2005;37:135–9.
17. Bouso JC, Andión Ó, Sarris JJ, Scheidegger M, Tófoli LF, Opaleye ES, et al. Adverse effects of ayahuasca: results from the global Ayahuasca survey. *PLoS Glob Public Health*. 2022;2:e0000438.
18. White E, Kennedy T, Ruffell S, Perkins D, Sarris J. Ayahuasca and dimethyltryptamine adverse events and toxicity analysis: a systematic thematic review. *Int J Toxicol*. 2024;43:327–39.
19. Colaço CS, Alves SS, Nolli LM, Pinheiro WO, de Oliveira DGR, Santos BWL, et al. Toxicity of ayahuasca after 28 days daily exposure and effects on monoamines and brain-derived neurotrophic factor (BDNF) in brain of Wistar rats. *Metab Brain Dis*. 2020;35:739–51.
20. Daldegan-Bueno D, Simionato NM, Favaro VM, Maia LO. The current state of ayahuasca research in animal models: a systematic review. *Prog Neuropsychopharmacol Biol Psychiatry*. 2023;125:110738.
21. da Motta LG, de Moraes JA, Tavares ACAM, Vianna LMS, Mortari MR, Amorim RFB, et al. Maternal and developmental toxicity of the hallucinogenic plant-based beverage ayahuasca in rats. *Reprod Toxicol*. 2018;77:143–53.
22. Pic-Taylor A, da Motta LG, de Moraes JA, Junior WM, Santos Ae, Campos F. Behavioural and neurotoxic effects of Ayahuasca infusion (Banisteriopsis Caapi and psychotria viridis) in female Wistar rat. *Behav Processes*. 2015;118:102–10.
23. Nolli LM, de Oliveira DGR, Alves SS, von Zuben MV, Pic-Taylor A, Mortari MR, et al. Effects of the hallucinogenic beverage ayahuasca on voluntary ethanol intake by rats and on cFos expression in brain areas relevant to drug addiction. *Alcohol*. 2020;84:67–75.
24. Acharezzi B, Tangerino GC, Sperandio LG, Mestriner ACD, Malfará WR. Hepatic evaluation in Wistar rats exposed to Ayahuasca tea. 2015; Available from: <https://www.cabidigitallibrary.org/doi/full/https://doi.org/10.5555/20163092787>
25. Durante Í, Dos Santos RG, Bouso JC, Hallak JE. Risk assessment of Ayahuasca use in a religious context: self-reported risk factors and adverse effects. *Braz J Psychiatry*. 2021;43:362–9.
26. Callaway JC, Grob CS. Ayahuasca preparations and serotonin reuptake inhibitors: a potential combination for severe adverse interactions. *J Psychoactive Drugs*. 1998;30:367–9.
27. Halman A, Kong G, Sarris J, Perkins D. Drug-drug interactions involving classic psychedelics: a systematic review. *J Psychopharmacol*. 2024;38:3–18.
28. Pontual AADD, Tófoli LF, Collares CF, Ramaekers JG, Corradi-Webster CM. The setting questionnaire for the Ayahuasca experience: questionnaire development and internal structure. *Front Psychol*. 2021;12:2347.
29. Diamant M, Gomer BR, Tófoli LF. In: Beatriz Labate CC, editor. Ayahuasca and psychotherapy: beyond integration. Ayahuasca Healing and Science. 2019;1:1–249.
30. Marinho EAV, Serra YA, Oliveira-Lima AJ, Marcourakis T, Berro LF. Ayahuasca for the treatment of alcohol use disorder. *Int Rev Neurobiol*. 2024;178:283–300.
31. Gianfratti B, Tabach R, Sakalem ME, Stessuk T, Maia LO, Carlini EA. Ayahuasca blocks ethanol preference in an animal model of dependence and shows no acute toxicity. *J Ethnopharmacol*. 2022;285:114865.
32. Serra YA, Barros-Santos T, Anjos-Santos A, Kisaki ND, Jovita-Farias C, Leite JPC, et al. Role of 5-HT_{2A} receptors in the effects of ayahuasca on ethanol self-administration using a two-bottle choice paradigm in male mice. *Psychopharmacology (Berl)*. 2022;239:1679–87.
33. Almeida CAF, Pereira-Junior AA, Rangel JG, Pereira BP, Costa KCM, Bruno V, et al. Ayahuasca, a psychedelic beverage, modulates neuroplasticity induced by ethanol in mice. *Behav Brain Res*. 2022;416:113546.
34. Reis HS, Rodrigues IRS, Anjos-Santos A, Libarino-Santos M, Serra YA, Cata-Preta EG, et al. Ayahuasca blocks the reinstatement of methylphenidate-induced conditioned place preference in mice: behavioral and brain Fos expression evaluations. *Psychopharmacology (Berl)*. 2020;237:3269–81.
35. Kisaki ND, Serra YA, Rodrigues IRS, Silva KSO, Jovita-Farias C, Alves GL, et al. Ayahuasca drinking using a two-bottle choice procedure in male mice. *Sci Rep*. 2024;14:25064.
36. Rodrigues LS, Rossi GN, Rocha JM, L Osório F, Bouso JC, Hallak JEC, et al. Effects of ayahuasca and its alkaloids on substance use disorders: an updated (2016–2020) systematic review of pre-clinical and human studies. *Eur Arch Psychiatry Clin Neurosci*. 2022;272:541–56.
37. Hamill J, Hallak J, Dursun SM, Baker G, Ayahuasca. Psychological And Physiologic Effects, Pharmacology And Potential Uses In Addiction And Mental Illness. *Curr Neuropharmacol [Internet]*. 2018; Available from: <https://doi.org/10.2174/1570159X16666180125095902>
38. Lawn W, Hallak JE, Crippa JA, Dos Santos R, Porffy L, Barratt MJ, et al. Well-being, problematic alcohol consumption and acute subjective drug effects in past-year ayahuasca users: a large, international, self-selecting online survey. *Sci Rep*. 2017;7:15201.
39. Perkins D, Sarris J, Cowley-Court T, Aicher H, Tófoli LF, Carlos Bouso J et al. Associations between Ayahuasca use in naturalistic settings and mental health and wellbeing outcomes: analysis of a large global dataset. *J Psychoact Drugs*. 2024;1:1–12.
40. Barbosa PC, Tofoli LF, Bogenschütz MP, Hoy R, Berro LF, Marinho EAV, et al. Assessment of alcohol and tobacco use disorders among religious users of Ayahuasca. *Front Psychiatry*. 2018;9:136.
41. Perkins D, Pagni BA, Sarris J, Barbosa PCR, Chenhall R. Changes in mental health, wellbeing and personality following ayahuasca consumption: results of a naturalistic longitudinal study. *Front Pharmacol*. 2022;13:884703.
42. Berlowitz I, Egger K, Cumming P. Monoamine oxidase inhibition by plant-derived β -carbolines; implications for the

- psychopharmacology of tobacco and ayahuasca. *Front Pharmacol.* 2022;13:886408.
43. Doering-Silveira E, Grob CS, de Rios MD, Lopez E, Alonso LK, Tacla C, et al. Report on psychoactive drug use among adolescents using ayahuasca within a religious context. *J Psychoactive Drugs.* 2005;37:141–4.
 44. Maia LO, Daldegan-Bueno D, Wießner I, Araujo DB, Tófoli LF. Ayahuasca's therapeutic potential: what we know - and what not. *Eur Neuropsychopharmacol.* 2023;66:45–61.
 45. Rush B, Marcus O, García S, Loizaga-Velder A, Spitalier A, Mendive F. Ayahuasca treatment outcome project (ATOP): one-year results from Takiwasi center and implications for psychedelic science. *J Stud Alcohol Drugs.* 2024;85:607–18.
 46. Daldegan-Bueno D, Maia LO, Massarenti CM, Tófoli LF. Ayahuasca and tobacco smoking cessation: results from an online survey in Brazil. *Psychopharmacology.* 2022;239:1767–82.
 47. Sheth R, Parikh E, Olayeye K, Pfeifer K, Khanna D. The effects of Ayahuasca on psychological disorders: a systematic literature review. *Cureus.* 2024;16:e55574.
 48. de Castro-Neto EF, da Cunha RH, da Silveira DX, Yonamine M, Gouveia TLF, Cavalheiro EA, et al. Changes in aminoacidergic and monoaminergic neurotransmission in the hippocampus and amygdala of rats after ayahuasca ingestion. *World J Biol Chem.* 2013;4:141–7.
 49. de Lima Osório F, de Macedo LRH, de Sousa JPM, Pinto JP, Quevedo J, de Souza Crippa JA et al. 5 The therapeutic potential of Harmine and Ayahuasca in depression: evidence from exploratory animal and human studies. *Ethnopharmacol Ayahuasca.* 2011;1:75–85.
 50. Fortunato JJ, Réus GZ, Kirsch TR, Stringari RB, Stertz L, Kapczinski F, et al. Acute harmine administration induces antidepressant-like effects and increases BDNF levels in the rat hippocampus. *Prog Neuropsychopharmacol Biol Psychiatry.* 2009;33:1425–30.
 51. Fortunato JJ, Réus GZ, Kirsch TR, Stringari RB, Fries GR, Kapczinski F, et al. Effects of beta-carboline harmine on behavioral and physiological parameters observed in the chronic mild stress model: further evidence of antidepressant properties. *Brain Res Bull.* 2010;81:491–6.
 52. Fortunato JJ, Réus GZ, Kirsch TR, Stringari RB, Fries GR, Kapczinski F, et al. Chronic administration of harmine elicits antidepressant-like effects and increases BDNF levels in rat hippocampus. *J Neural Transm.* 2010;117:1131–7.
 53. Zheng Z-H, Lin X-C, Lu Y, Cao S-R, Liu X-K, Lin D, et al. Harmine exerts anxiolytic effects by regulating neuroinflammation and neuronal plasticity in the basolateral amygdala. *Int Immunopharmacol.* 2023;119:110208.
 54. daSilva FS, Silva EAS, de Sousa GM Jr, Maia-de-Oliveira JP, Soares-Rachetti V, de Araujo P, DB. Acute effects of Ayahuasca in a juvenile non-human primate model of depression. *Braz J Psychiatry.* 2018;1:280–8. <https://doi.org/10.1590/1516-4446-2018-0140>
 55. de Meiroz Grilo MLP, de Sousa GM, de Mendonça LAC, Lobão-Soares B, de Sousa MBC, Palhano-Fontes F, et al. Prophylactic action of Ayahuasca in a non-human primate model of depressive-like behavior. *Front Behav Neurosci.* 2022;16:901425.
 56. de Lira-Bandeira WG, de Mendonça Batista LAC, de Medeiros Bandeira AS, de Góis Morais PLA, Pereira LRF, de Meiroz Grilo MLP, et al. Ayahuasca partially preserves striatal integrity in juvenile non-human primates exposed to chronic stress: evidence from Stereological evaluation. *Front Neuroanat.* 2025;19:1457557.
 57. Ruffell S, Netzband N, Bird C, Young AH, Juruena MF. The pharmacological interaction of compounds in ayahuasca: a systematic review. *Rev Bras Psiquiatr.* 2020;42:646–56.
 58. de Sousa GM, de Oliveira Tavares VD, de Menezes Galvão AC, de Almeida RN, Palhano-Fontes F, Lobão-Soares B, et al. Moderators of ayahuasca's biological antidepressant action. *Front Psychiatry.* 2022;13:1033816.
 59. Xavier J, Farias CP, Soares MS, Silveira GO, Spanevello R, Yonamine M, et al. Ayahuasca prevents oxidative stress in a rat model of depression elicited by unpredictable chronic mild stress. *Revista De Psiquiatria Clin.* 2021;48:90–8.
 60. Liao C, Dua AN, Wojtasiewicz C, Liston C, Kwan AC. Structural neural plasticity evoked by rapid-acting antidepressant interventions. *Nat Rev Neurosci.* 2025;26:101–14.
 61. Labate BC, Cavnar C. The therapeutic use of Ayahuasca. In: Labate BC, Cavnar C, editors. Vol. 1. Springer Berlin Heidelberg; 2013. pp. 1–226.
 62. Barbosa PC, Mizumoto S, Bogenschutz MP, Strassman RJ. Health status of Ayahuasca users. *Drug Test Anal.* 2012;4:601–9.
 63. Barbosa PCR, Giglio JS, Dalgalarondo P. Altered states of consciousness and short-term psychological after-effects induced by the first time ritual use of Ayahuasca in an urban context in Brazil. *J Psychoact Drugs.* 2005;37:193–201.
 64. Santos RG, Landeira-Fernandez J, Strassman RJ, Motta V, Cruz APM. Effects of Ayahuasca on psychometric measures of anxiety, panic-like and hopelessness in Santo Daime members. *J Ethnopharmacol.* 2007;112:507–13.
 65. Sarris J, Perkins D, Cribb L, Schubert V, Opaleye E, Bouso JC, et al. Ayahuasca use and reported effects on depression and anxiety symptoms: an international cross-sectional study of 11,912 consumers. *J Affect Disord Rep.* 2021;4:100098.
 66. Ruffell SGD, Netzband N, Tsang W, Davies M, Butler M, Rucker JJH, et al. Ceremonial Ayahuasca in Amazonian retreats-mental health and epigenetic outcomes from a six-month naturalistic study. *Front Psychiatry.* 2021;12:687615.
 67. Uthaug MV, Mason NL, Toennes SW, Reckweg JT, de Sousa Fernandes Perna EB, Kuypers KPC, et al. A placebo-controlled study of the effects of Ayahuasca, set and setting on mental health of participants in Ayahuasca group retreats. *Psychopharmacology.* 2021;238:1899–910.
 68. Osório F, Sanches RF, Macedo LR, dos Santos RG, Maia-de-Oliveira JP, Wichert-Ana L, et al. Antidepressant effects of a single dose of ayahuasca in patients with recurrent depression: a preliminary report. *Rev Bras Psiquiatr.* 2015;37:13–20.
 69. Sanches RF, de Lima Osório F, dos Santos RG, Macedo LRH, Maia-de-Oliveira JP, Wichert-Ana L, et al. Antidepressant effects of a single dose of Ayahuasca in patients with recurrent depression. *J Clin Psychopharmacol.* 2016;36:77–81.
 70. Palhano-Fontes F, Barreto D, Onias H, Andrade KC, Novaes MM, Pessoa JA, et al. Rapid antidepressant effects of the psychedelic Ayahuasca in treatment-resistant depression: a randomized placebo-controlled trial. *Psychol Med.* 2019;49:655–63.
 71. Zeifman R, Palhano-Fontes F, Hallak J. The Impact of Ayahuasca on Suicidality: Results from a Randomized Controlled Trial. *Frontiers in [Internet].* 2019; Available from: <https://www.frontiersin.org/articles/https://doi.org/10.3389/fphar.2019.01325>
 72. Strassman RJ, Qualls CR, Uhlenhuth EH, Kellner R. Dose-response study of N,N-dimethyltryptamine in humans: II. Subjective effects and preliminary results of a new rating scale. *Arch Gen Psychiatry.* 1994;51:98–108.
 73. Falchi-Carvalho M, Wießner I, Silva SRB, O Maia L, Barros H, Laborde S, et al. Safety and tolerability of inhaled N,N-Dimethyltryptamine (BMND01 candidate): A phase I clinical trial. *Eur Neuropsychopharmacol.* 2024;80:27–35.
 74. Falchi-Carvalho M, Barros H, Bolcont R, Laborde S, Wießner I, Ruschi B, Silva S et al. The antidepressant effects of vaporized N,N-dimethyltryptamine: An open-label pilot trial in treatment-resistant depression. *Psychodelic Medicine [Internet].* 2024; Available from: <https://doi.org/10.1089/psymed.2024.0002>

75. Timmermann C, Zeifman RJ, Erritzoe D, Nutt DJ, Carhart-Harris RL. Effects of DMT on mental health outcomes in healthy volunteers. *Sci Rep*. 2024;14:3097.
76. D'Souza DC, Syed SA, Flynn LT, Safi-Aghdam H, Cozzi NV, Ranganathan M. Exploratory study of the dose-related safety, tolerability, and efficacy of dimethyltryptamine (DMT) in healthy volunteers and major depressive disorder. *Neuropsychopharmacology*. 2022. <https://doi.org/10.1038/s41386-022-01344-y>.
77. Falchi-Carvalho M, Palhano-Fontes F, Wießner I, Barros H, Bolcont R, Laborde S, et al. Rapid and sustained antidepressant effects of vaporized N,N-dimethyltryptamine: a phase 2a clinical trial in treatment-resistant depression. *Neuropsychopharmacology*. 2025;50:895–903.
78. Daws RE, Timmermann C, Giribaldi B, Sexton JD, Wall MB, Erritzoe D, et al. Increased global integration in the brain after psilocybin therapy for depression. *Nat Med*. 2022;28:844–51.
79. Morales-Garcia JA, Calleja-Conde J, Lopez-Moreno JA, Alonso-Gil S, Sanz-SanCristobal M, Riba J, et al. N,N-dimethyltryptamine compound found in the hallucinogenic tea ayahuasca, regulates adult neurogenesis in vitro and in vivo. *Transl Psychiatry*. 2020;10:331.
80. Moliner R, Giryh M, Brunello CA, Kovaleva V, Biojone C, Enkavi G, et al. Psychedelics promote plasticity by directly binding to BDNF receptor TrkB. *Nat Neurosci*. 2023;26:1032–41.
81. Werle I, Bertoglio LJ. Psychedelics: a review of their effects on recalled aversive memories and fear/anxiety expression in rodents. *Neurosci Biobehav Rev*. 2024;167:105899.
82. Daneluz DM, Sohn JMB, Silveira GO, Yonamine M, Stern CA. Evidence on the impairing effects of Ayahuasca on fear memory reconsolidation. *Psychopharmacology*. 2022;239:3325–36.
83. Nielson JL, Megler JD, Cavnar C. A qualitative assessment of risks and benefits of ayahuasca for trauma survivors. In: *Ayahuasca Healing and Science*. Cham: Springer International Publishing; 2021. p. 81–98.
84. Weiss B, Dinh-Williams L-AL, Beller N, Raugh IM, Strauss GP, Campbell WK. Ayahuasca in the treatment of posttraumatic stress disorder: Mixed-methods case series evaluation in military combat veterans. *Psychol Trauma* [Internet]. 2023; Available from: <https://doi.org/10.1037/tra0001625>
85. Savoldi R, Polari D, Pinheiro-da-Silva J, Silva PF, Lobao-Soares B, Yonamine M, et al. Behavioral changes over time following Ayahuasca exposure in zebrafish. *Front Behav Neurosci*. 2017;11:139.
86. da Goulart Silva M, Daros GC, Santos FP, Yonamine M, de Bitencourt RM. Antidepressant and anxiolytic-like effects of ayahuasca in rats subjected to LPS-induced neuroinflammation. *Behav Brain Res*. 2022;434:114007.
87. Hilber P, Chapillon P. Effects of harmaline on anxiety-related behavior in mice. *Physiol Behav*. 2005;86:164–7.
88. de Camargo RW, Joaquim L, Machado RS, de Souza Ramos S, da Rosa LR, de Novais Junior LR et al. Ayahuasca pretreatment prevents sepsis-induced anxiety-like behavior, neuroinflammation, and oxidative stress, and increases brain-derived neurotrophic factor. *Mol Neurobiol* [Internet]. 2024; Available from: https://link.springer.com/authorize/casa?redirect_uri=https://link.springer.com/article/https://doi.org/10.1007/s12035-024-04597-4&casa_token=l6gCDeWKYu0AAAAA:S36zDHUMLXxtVa_-jdhXaCMjgRkDFcaUbSVACqh6MoDK4Vun8XOVFzvFqk-Z5bR-uvRyITxnKXTz17Z81Q.
89. Silva MS, Lima L, Saraiva K, Correa R, Cardoso AP, Silva J. O uso terapêutico da Ayahuasca no transtorno de ansiedade generalizada: uma revisão sistemática da literatura. *OBSERVATÓRIO DE LA ECONOMÍA LATINOAMERICANA*. 2024;22:e5671.
90. Dos Santos RG, Osório F, de Rocha L, Rossi JM, Bouso GN, Rodrigues JC. Ayahuasca improves self-perception of speech performance in subjects with social anxiety disorder: A pilot, proof-of-concept, randomized, placebo-controlled trial. *J Clin Psychopharmacol*. 2021;41:540–50.
91. Renelli M, Fletcher J, Tupper KW, Files N, Loizaga-Velder A, Lafrance A. An exploratory study of experiences with conventional eating disorder treatment and ceremonial ayahuasca for the healing of eating disorders. *Eat Weight Disord*. 2018. <https://doi.org/10.1007/s40519-018-0619-6>.
92. Lafrance A, Renelli M, Fletcher J, Files N, Tupper KW, Loizaga-Velder A. Ayahuasca as a healing tool along the continuum of recovery from eating disorders. In: *Ayahuasca Healing and Science*. Cham: Springer International Publishing; 2021. p. 189–208.
93. Calder A, Mock S, Friedli N, Pasi P, Hasler G. Psychedelics in the treatment of eating disorders: rationale and potential mechanisms. *Eur Neuropsychopharmacol*. 2023;75:1–14.
94. Spriggs MJ, Kettner H, Carhart-Harris RL. Positive effects of psychedelics on depression and wellbeing scores in individuals reporting an eating disorder. *Eat Weight Disord*. 2021;26:1265–70.
95. Hu S, Lin C, Wang H, Wang X. Psychedelics and eating disorders: Exploring the therapeutic potential for anorexia nervosa and beyond. *ACS Pharmacol Transl Sci* [Internet]. 2025; Available from: <https://pubs.acs.org/doi/abs/https://doi.org/10.1021/acspsc.5c00094>
96. Lafrance A, Loizaga-Velder A, Fletcher J, Renelli M, Files N, Tupper KW. Nourishing the spirit: exploratory research on Ayahuasca experiences along the continuum of recovery from eating disorders. *J Psychoact Drugs*. 2017;49:427–35.
97. Graziosi M, Rohde JS, Tiwari P, Siev J, Yaden DB. Psychedelics, OCD and related disorders: a systematic review. *J Obsessive Compuls Relat Disord*. 2024;41:100873.
98. Brewerton TD, Wang JB, Lafrance A, Pamplin C, Mithoefer M, Yazar-Klosinski B, et al. MDMA-assisted therapy significantly reduces eating disorder symptoms in a randomized placebo-controlled trial of adults with severe PTSD. *J Psychiatr Res*. 2022;149:128–35.
99. Williams M, Kingston Miller A, Loizaga-Velder A, Files N, Lafrance A. Getting to the root: Ayahuasca ceremony leaders' perspectives on eating disorders. *J Psychoact Drugs*. 2023;55:425–33.
100. Carrithers BM, Roberts DE, Weiss BM, King JD, Carhart-Harris RL, Gordon AR, et al. Exploring serotonergic psychedelics as a treatment for personality disorders. *Neuropharmacology*. 2025;272:110413.
101. Mendes Rocha J, Rossi GN, Osório FL, Bouso Saiz JC, Silveira GDO, Yonamine M, et al. Effects of ayahuasca on personality: results of two randomized, placebo-controlled trials in healthy volunteers. *Front Psychiatry*. 2021;12:688439.
102. Domínguez-Clavé E, Soler J, Pascual JC, Elices M, Franquesa A, Valle M, et al. Ayahuasca improves emotion dysregulation in a community sample and in individuals with borderline-like traits. *Psychopharmacology*. 2019;236:573–80.
103. Sampedro F, de la Fuente Revenga M, Valle M, Roberto N, Domínguez-Clavé E, Elices M, et al. Assessing the psychedelic “after-glow” in Ayahuasca users: post-acute neurometabolic and functional connectivity changes are associated with enhanced mindfulness capacities. *Int J Neuropsychopharmacol*. 2017;20:698–711.
104. Bouso JC, Palhano-Fontes F, Rodríguez-Fornells A, Ribeiro S, Sanches R, Crippa JAS, et al. Long-term use of psychedelic drugs is associated with differences in brain structure and personality in humans. *Eur Neuropsychopharmacol*. 2015;25:483–92.
105. Netzbund N, Ruffell S, Linton S, Tsang WF, Wolff T. Modulatory effects of Ayahuasca on personality structure in a traditional framework. *Psychopharmacology*. 2020;237:3161–71.
106. Weiss BM, Miller JD, Carter NT, Campbell WK. Examining changes in personality following shamanic ceremonial use of ayahuasca. *Sci Rep* [Internet]. 2020;11. Available from: <https://www.nature.com/articles/s41598-021-84746-0>

107. Arruda Sanchez T, Ramos LR, Araujo F, Schenberg EE, Yonamine M, Lobo I, et al. Emotion regulation effects of Ayahuasca in experienced subjects during implicit aversive stimulation: an fMRI study. *J Ethnopharmacol*. 2024;320:117430.
108. Koslowski M, Gasser P. Guiding through challenging psychedelic experiences and bad trips. *Curr Top Behav Neurosci* [Internet]. 2024; Available from: https://link.springer.com/chapter/10.1007/7854_2024_515
109. Kočárová R, Horáček J, Carhart-Harris R. Does psychedelic therapy have a transdiagnostic action and prophylactic potential? *Front Psychiatry*. 2021;12:661233.
110. de Almeida RN, Galvão AC, de Silva M, Silva FS, Palhano-Fontes EADS, Maia-de-Oliveira F. Modulation of serum Brain-Derived neurotrophic factor by a single dose of ayahuasca: observation from a randomized controlled trial. *Front Psychol*. 2019;10:1234.
111. Galvão-Coelho NL, de Menezes Galvão AC, de Almeida RN, Palhano-Fontes F, Campos Braga I, Lobão Soares B, et al. Changes in inflammatory biomarkers are related to the antidepressant effects of Ayahuasca. *J Psychopharmacol*. 2020. <https://doi.org/10.1177/0269881120936486>.
112. Palhano-Fontes F, Soares BL, Galvão-Coelho NL, Arcoverde E, Araujo DB. Ayahuasca for the treatment of depression. *Curr Top Behav Neurosci*. 2022;56:113–24.
113. Palhano-Fontes F, Andrade KC, Tofoli LF, Jose ACS, Crippa AS, Hallak JEC et al. The psychedelic state induced by Ayahuasca modulates the activity and connectivity of the Default Mode Network. *PLoS One* [Internet]. 2015;10. Available from: <https://doi.org/10.1371/journal.pone.0118143>
114. Pasquini L, Palhano-Fontes F, Araujo DB. Subacute effects of the psychedelic ayahuasca on the salience and default mode networks. *J Psychopharmacol*. 2020. <https://doi.org/10.1177/0269881120909409>.
115. De Araujo DB, Ribeiro S, Cecchi GA, Carvalho FM, Sanchez TA, Pinto JP, et al. Seeing with the eyes shut: neural basis of enhanced imagery following Ayahuasca ingestion. *Hum Brain Mapp*. 2012;33:2550–60.
116. Franquesa A, Sainz-Cort A, Gandy S, Soler J, Alcázar-Córcoles MÁ, Bouso JC. Psychological variables implied in the therapeutic effect of ayahuasca: a contextual approach. *Psychiatry Res*. 2018;264:334–9.
117. Romeo B, Kervadec E, Fauvel B, Strika-Bruneau L, Amirouche A, Bezo A, et al. The intensity of the psychedelic experience is reliably associated with clinical improvements: A systematic review and meta-analysis. *Neurosci Biobehav Rev*. 2025;172:106086.
118. Agin-Liebes G, Zeifman R, Luoma JB, Garland EL, Campbell WK, Weiss B. Prospective examination of the therapeutic role of psychological flexibility and cognitive reappraisal in the ceremonial use of ayahuasca. *J Psychopharmacol*. 2022;36:295–308.
119. Mian MN, Altman BR, Earleywine M. Ayahuasca's antidepressant effects covary with behavioral activation as well as mindfulness. *J Psychoact Drugs*. 2020;52:130–7.
120. Viol A, Palhano-Fontes F, Onias H, de Araujo DB, Viswanathan GM. Shannon entropy of brain functional complex networks under the influence of the psychedelic Ayahuasca. *Sci Rep*. 2017;7:7388.
121. Viol A, Viswanathan GM, Soldatkin O, Fontes FP, Onias H, de Araujo DB et al. Information parity increases on functional brain networks under influence of a psychedelic substance. *J Phys Complex* [Internet]. 2023 [cited 2023 Mar 11]; Available from: <https://iopscience.iop.org/article/https://doi.org/10.1088/2632-072X/acc22b/meta>
122. Inserra A. Hypothesis: The psychedelic Ayahuasca heals traumatic memories via a sigma 1 receptor-mediated epigenetic-mnemonic process. *Front Pharmacol*. 2018;9:330.
123. Dos Santos RG, Hallak JEC. Ayahuasca: pharmacology, safety, and therapeutic effects. *CNS Spectr*. 2024;1:1–9.
124. Omágua-Kambebe A, Labate BC, Ribeiro S. Psychedelic science and Indigenous shamanism: an urgent dialogue. *Nat Ment Health*. 2023;1:815–6.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.